

Applied Machine Learning Pneumonia Classification on Children's Chest X-Ray Images

Anneke Catherine Naseef

S6490662

Marcus Harald Olof Persson

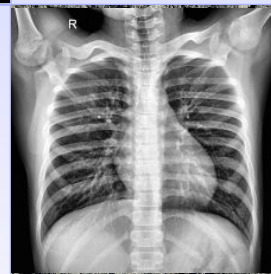
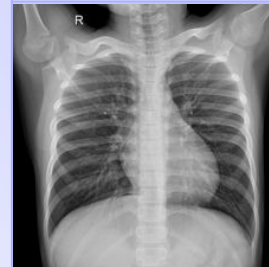
S5343798

Ignacio Jacob Uroz Rodríguez

S5118913

Julian Wilbert Sprietsma

S5096219



Agenda

01 Introduction

02 Preprocessing

03 Training & Results

04 Professionalism

05 Extras and Insights

06 Demo

Introduction

Context

- 1** Patient presents (pneumonia) symptoms.
- 2** Physician takes x-ray.
- 3** X-ray is inputted into application.
- 4** App outputs diagnosis/confidence measure.
- 5** Results used in diagnostics.

Novelty

1. From Binary to Distinction

- Typically, yes/no pneumonia.
- Distinguishing bacterial and viral pneumonia overlap allows for proper treatment.

3. Cheaper Applications

- Child pneumonia mortality is highest in low-resource areas.
- A cheaper solution that solves the same problem is beneficial.

2. Exploring Condensation

- Can we achieve as good of a performance with a smaller model?
- Not just train on raw images, but on image data.

4. Attacking Imbalance

- Bacterial pneumonia dominates our dataset.
- Augmentation and class weights allows for generalisation without new data.

Chosen Models

Baseline 3-layer CNN

What?

2 Convolution, 1 Fully Connected.

Why?

Serve as a baseline to measure to compare performance of models.

LightGBM as an Alternative

What?

Gradient-boosted decision tree model.

Why?

Explore if a fast, classical method can rival deep learning.

Pretrained ResNet-18

What?

Training on a new layer and classifier head.

Why?

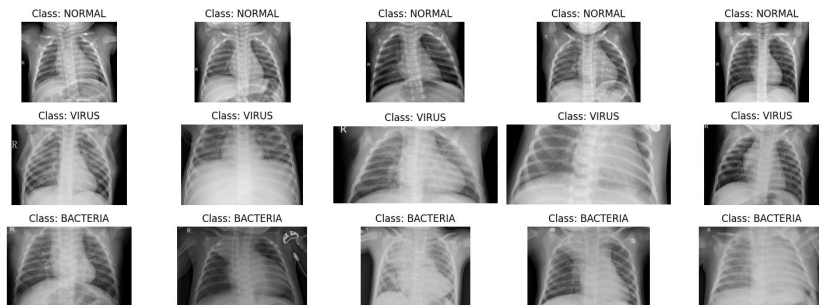
Leverage transfer learning from strong pretrained features.

Preprocessing

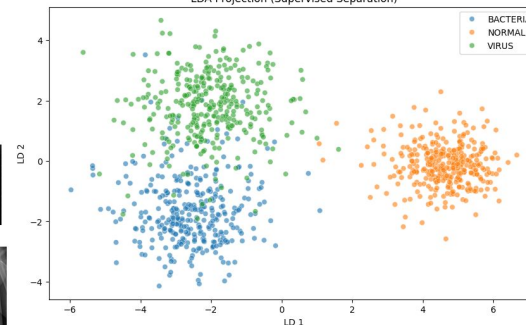
Preprocessing

Beautiful. Raw. Messy.

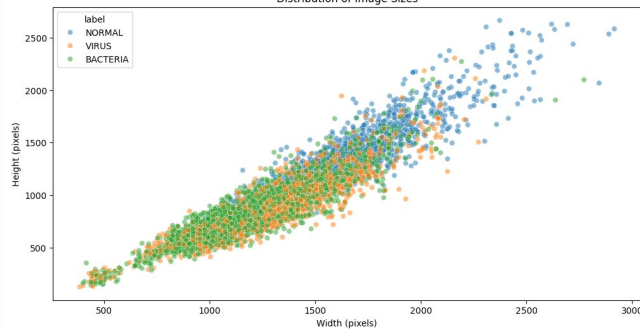
Sample Images from Each Class



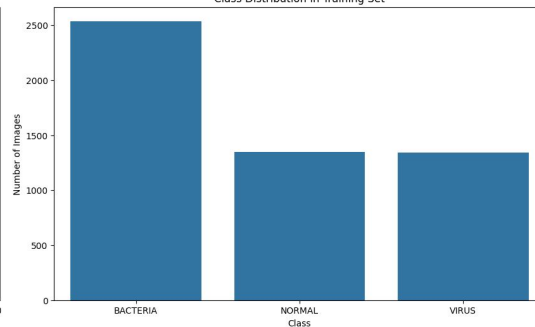
LDA Projection (Supervised Separation)



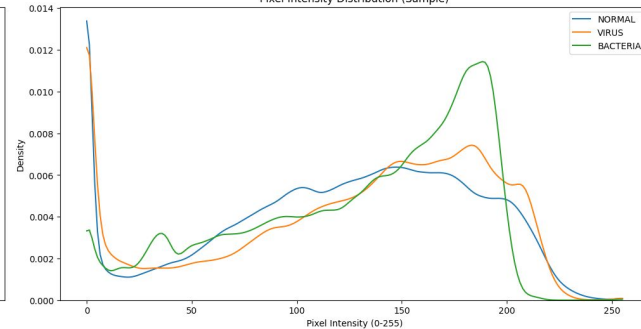
Distribution of Image Sizes



Class Distribution in Training Set



Pixel Intensity Distribution (Sample)



PyTorch Pipeline...

1. Lung Region Extraction

What? Contour mapping to isolate the lungs.

Why? Remove irrelevant borders and background for model focus.

2. Resize Standardisation

What? Rescale images to a fixed size.

Why? Address inconsistent dimensions, and requirement for uniform input in batching.

3. Contrast Enhancement

What? CLAHE to apply adaptive histogram equalisation.

Why? Normalise brightness/contrast differences from different X-rays.

4. Normalisation

What? Scale pixel values to a consistent range.

Why? Stabilization and training speedup.



Extended With the LightGBM Pipeline!

5. Downsampling

What? Shrink the processed image further.

Why? Keep tabular feature count small enough for a tree-based model.

6. Normalise to Numpy Array

What? Project pixel values into a $[0, 1]$ scaled array.

Why? Consistent scale for computing features.

7. Extract Features

What? Compute pixel statistics and HOG features.

Why? Capture structural texture at lower dimensionality instead of raw pixels.

Ending: Data Augmentation

What? Slight rotations, zooms, contrast shifts.

Why? Robustness and class balancing.

But What About Data Leakage?!

- **Assumption:** Kaggle dataset does not suffer from data leakage.
 - **Residual Risk:** Multiple x-rays of the same patient?
- **Issue:** Raw data is just train/test.
 - Train set is split 80/20 train-validation.
 - Validation set does not contain any augmented data.
- Preprocessing is done per image, not as a general whole.
- **Bonus:** LightGBM pipeline has both non-augmented and augmented datasets



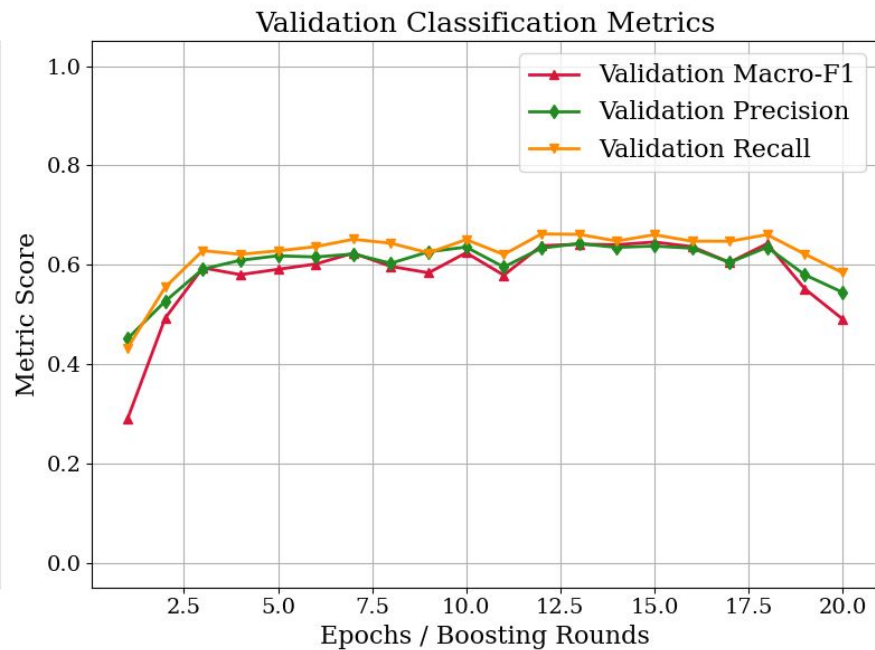
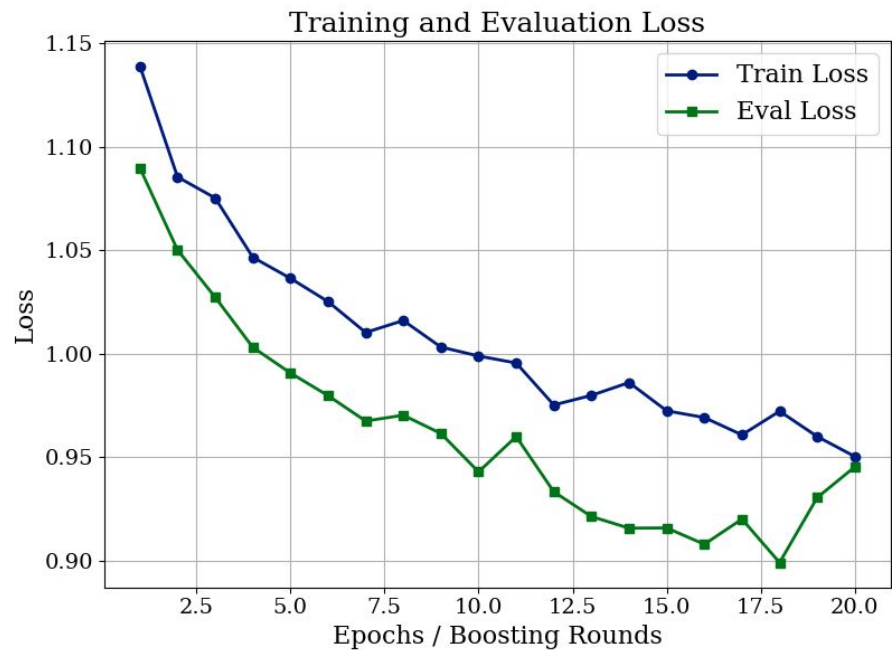
Training & Results

Hyperparameter Tuning & Training

- **Grid Search & 5-fold Stratified K-fold Cross-validation.**
- **Selection Metric:** Macro-F1 for tracking class balance.
- **Early Stopping:** 3 epochs.

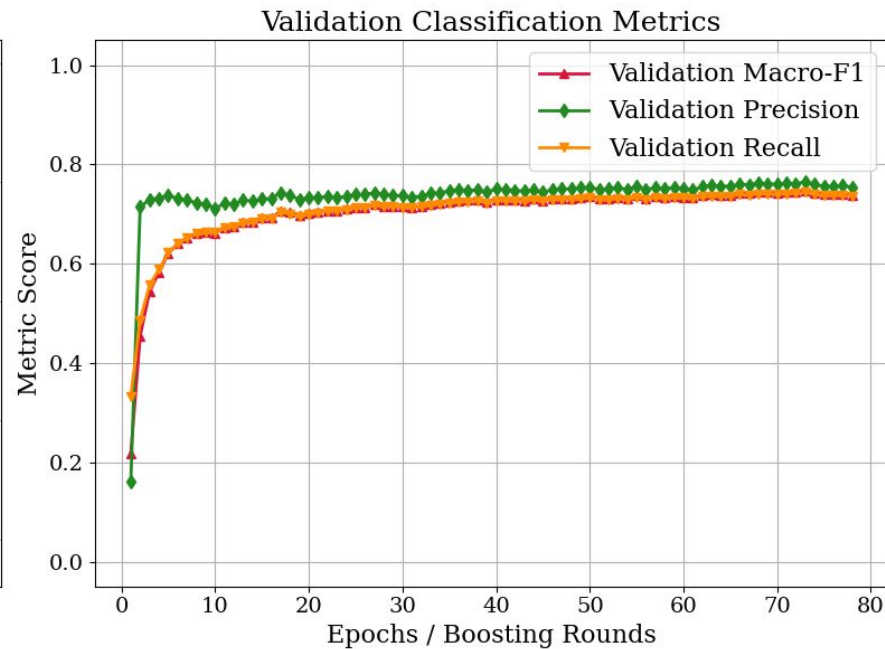
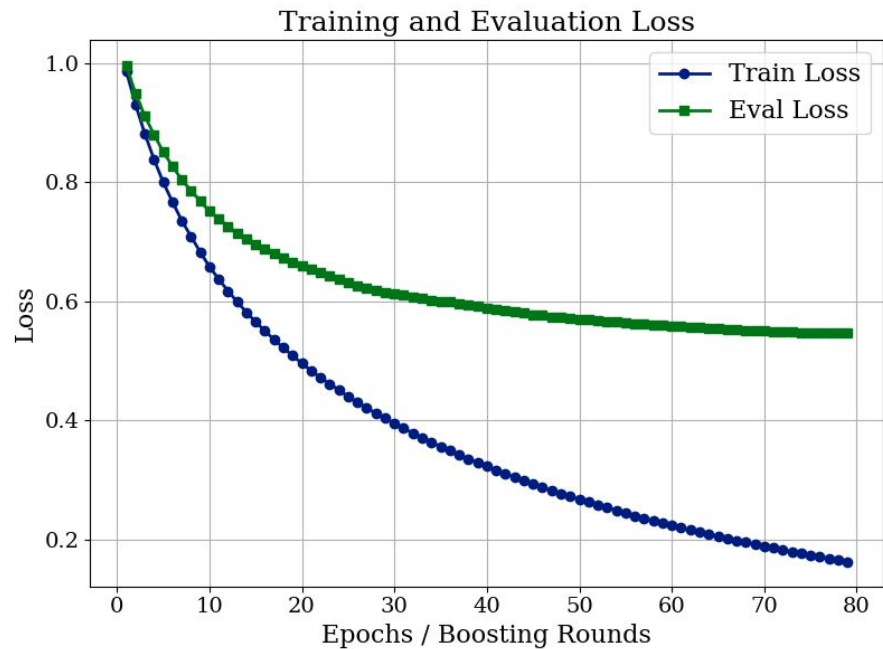
Model	Hyperparameters	Range
CNN/ResNet-18	Learning Rate	1e-4, 1e-3, 1e-2
	Batch Size	16, 32
	Weight Decay	0, 1e-4
LightGBM	Learning Rate	0.01, 0.05, 0.1
	Number of Leaves	15, 31, 63
	Maximum Depth	4, 6, Unlimited (-1)

CNN



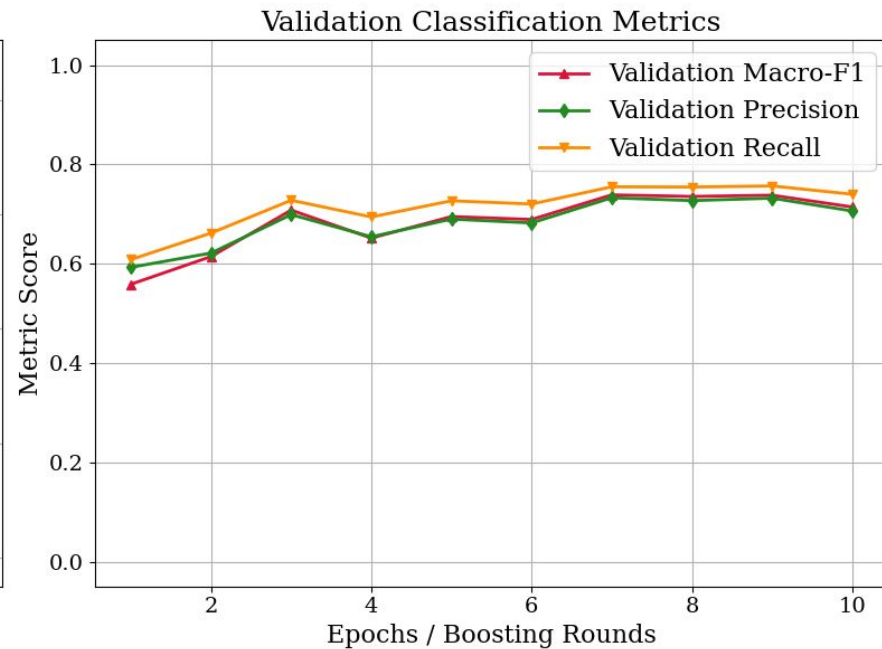
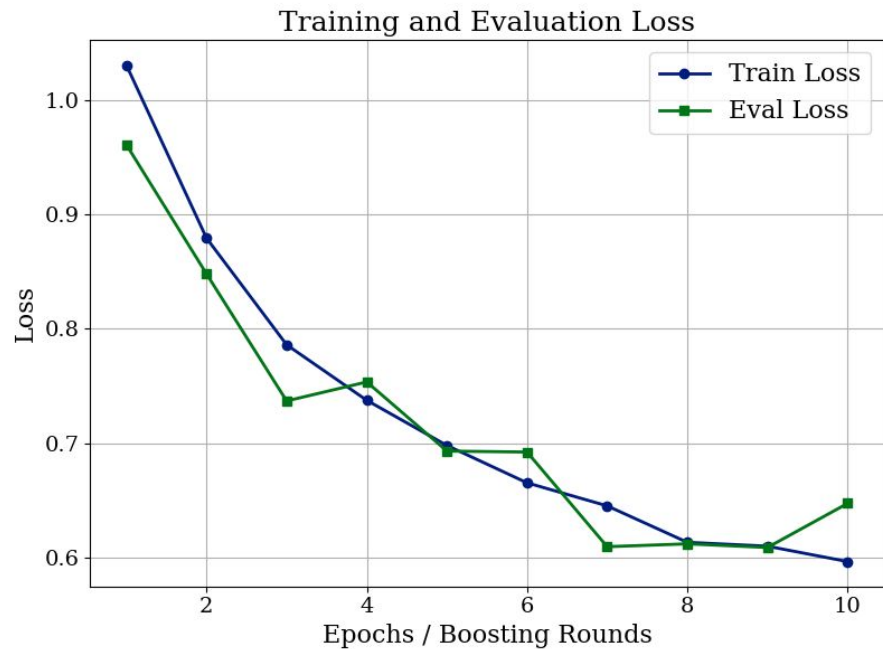
Learning Rate = 0.0001; Batch Size = 16; Weight Decay = 0.0001
Max Epochs = 20

LightGBM



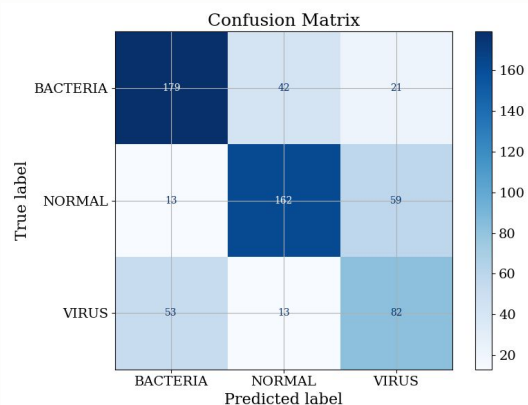
(Learning Rate = 0.1; Number of Leaves = 15; Maximum Depth = Infinite)
Max Epochs = 150

ResNet-18

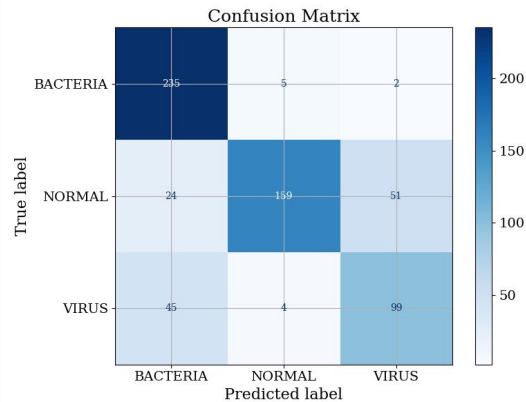


(Learning Rate = 0.0001; Batch Size = 32; Weight Decay = 0.0001)
Max Epochs = 10

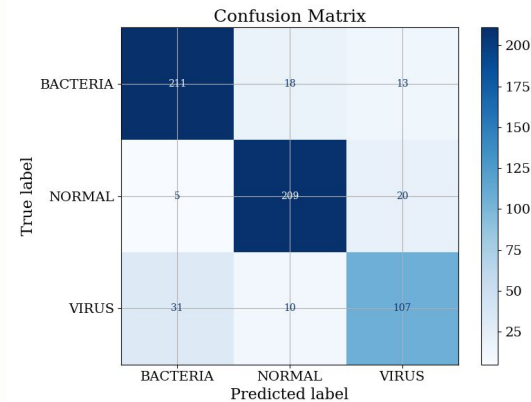
Confusion Matrices



CNN



LightGBM



ResNet-18

All three models confuse the two infectious classes and under-detect the minority VIRUS class; ResNet-18 best mitigates this, while LightGBM's high BACTERIA score reflects a majority-class bias rather than balanced skill, even after data augmentation.

professionalism



Professionalism

Fully Documented
Clear Code

Detailed README

Pre-Commit Hooks

uv Package
Management

Tensorboard for
Monitoring

Local Configs to Hide
Secrets

Logging Everywhere

Unit Tests With
GitHub Actions

Pull Request
Practices

Weekly Meetings at
the Friday Lab

Command Line
Interface (CLI)

Sync With Overleaf
for Quick Access

AML: Pneumonia Classification via Chest X-Rays

Repository for the Applied Machine Learning course (WBAI065-05) at the University of Groningen.

Running via Docker

1. Build the image.

```
docker compose build
```



2. Run the image.

```
docker compose up
```



This exposes the following ports:

- 8000 : FastAPI application.
- 8501 : Streamlit dashboard.
- 6006 : TensorBoard.

Once the stack is up, open the dashboard at <http://localhost:8501>.

Development

We use [uv](#) for project management.

1. Clone the project.
2. Synchronise the project.

```
uv sync
```

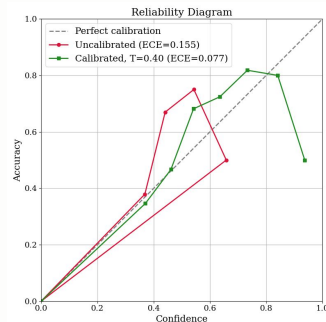
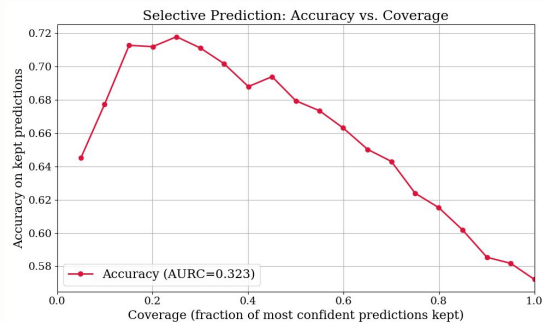


Extras and Insights

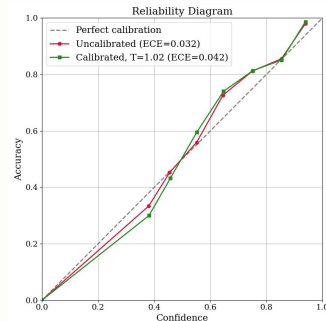
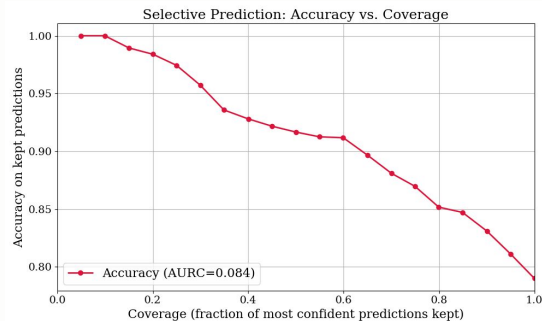
Uncertainty Quantification

- **Calibration:** Temperature scaling scored by ECE.
- **Selective Prediction:** Confidence-based Abstention (AURC).
- **Metrics:** ECE, Brier, NLL, Entropy
- **Most Underconfident:** CNN
- **Best Calibrated:** LightGBM
- **Best Selective Prediction:** ResNet-18

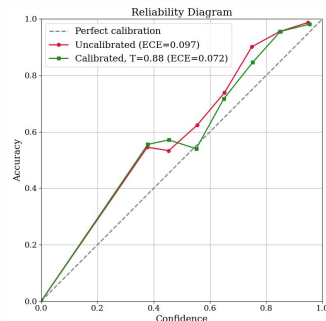
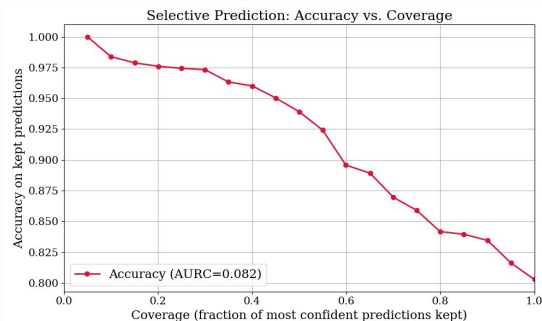
CNN



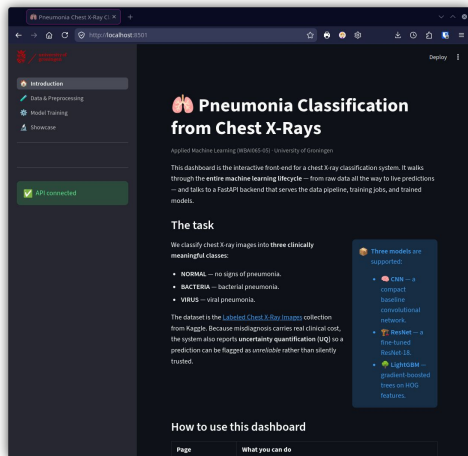
LightBM



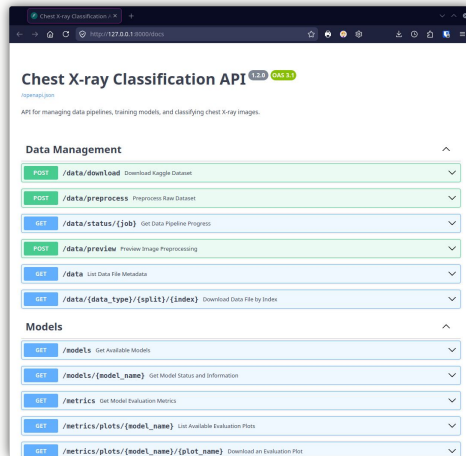
ResNet-18



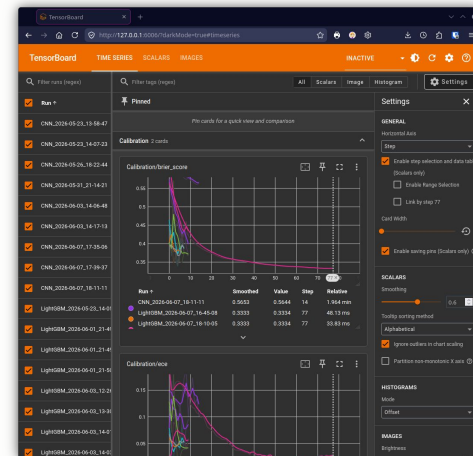
Containerisation



Frontend: Streamlit



Backend: FastAPI



Monitoring: Tensorboard



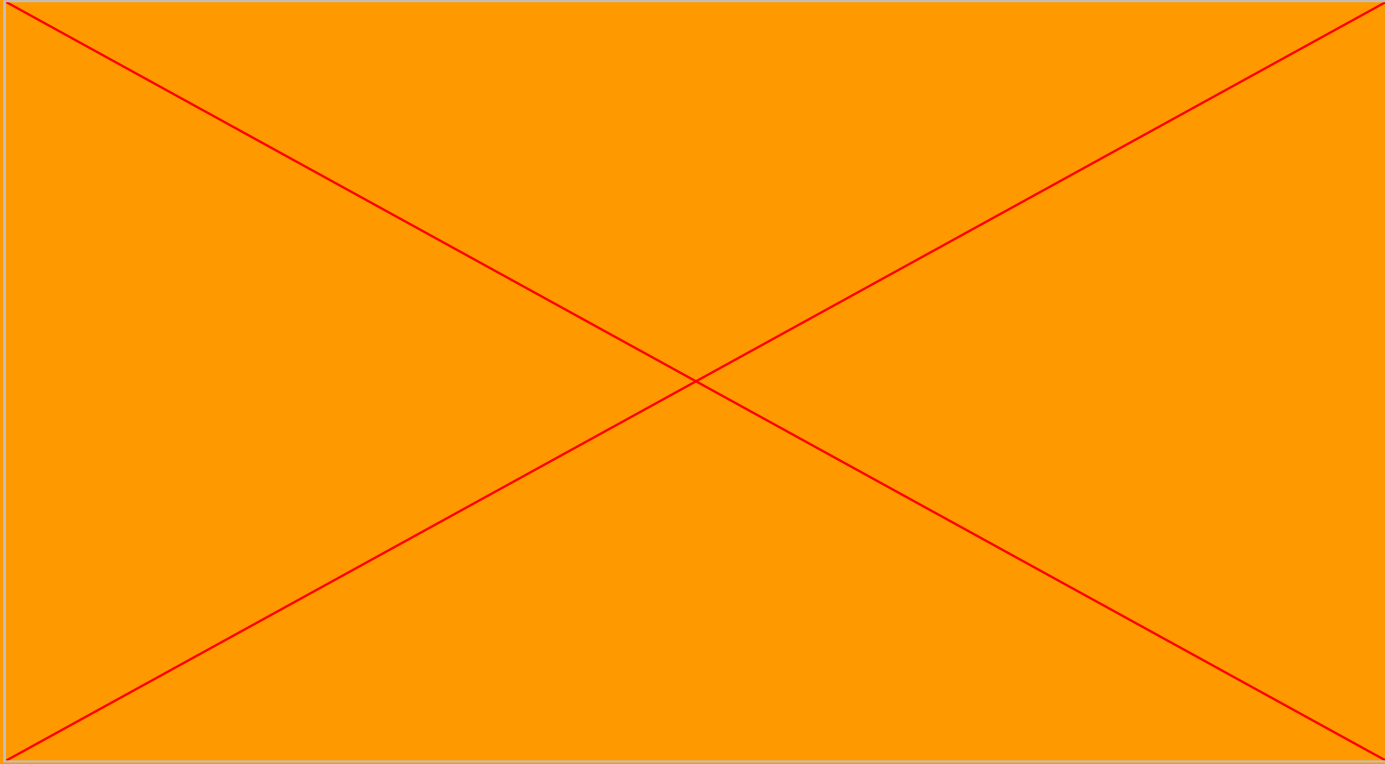
Insights

- 1** Bacterial against viral pneumonia is difficult from overlap.
- 2** We can frame “Pneumonia vs. Healthy” for 2 Stages.
- 3** Accuracy $\neq$ calibration.
- 4** VIRUS is the minority and bottleneck, capping F1.
- 5** The dataset is a big contributor.

Demo

Demo

Demo (Backup)



Thank You!



/anneke-c-naseef



/marcusper



/ignacio-uroz



/julian-sprietsma